



UNIVERSIDADE D
COIMBRA

PROJECT REPORT

BUSINESS INTELLIGENCE

Title: Wellness Tracker: Data-Driven Health & Fitness Insights

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Dataset

<https://www.kaggle.com/datasets/jijagallery/fitlife-health-and-fitness-tracking-dataset>

Description:

By analyzing demographic information, activity metrics, health indicators and lifestyle metrics the app aims to find connections between them and provide valuable insights that can contribute to the improvement of health and overall well-being.

Objectives/Requirements/Questions

1. for DW:

1. How do user workout trends (activity type) vary across different demographics (age, gender, BMI)?
2. What are the most effective activity types based on calories burned?
3. How does smoking affect health (stress level, sleep, duration and intensity of exercise)?
4. How demographic information can affect fitness performance (duration of exercise, steps count, intensity)?
5. What is the relationship between blood pressure (diastolic or systolic) and stress level?
6. What is the relationship between BMI and physical fitness, considering factors as exercise duration and calories burned?
7. What is the average number of steps per stress level and age group?
8. What is the relationship between stress levels and daily steps across different age groups?
9. How does hydration level affect the average heart rate during exercise?

2. for DM

1. Can we predict which users are most likely to stop tracking their fitness data based on past behavior? / Can we detect patterns in users who stop tracking their data based on stress levels and sleep habits?
2. How does season change affect user engagement and activity levels (duration in minutes, daily steps, calories burned)?
3. How can we cluster users based on workout intensity, frequency, and preferences?
4. Can we recommend personalized workout plans based on user similarities and historical data?
5. What percentage of users consistently track their health indicators, and how many stop the exercise (shorter duration of exercising) based on health feedback?
6. Can blood pressure history be used to predict the likelihood of reduced physical activity?
7. Can we predict the average workout duration of a user based on weight, age and fitness level?
8. How does the relationship between calories burned and workout intensity change by season?
9. How can we optimize personalized workout recommendations based on BMI and stress levels?

Customers:

Health and Wellness Centers, Fitness Apps

Main challenges:

Getting a loyal customer group

Expanding to a new market

The competition's price of similar services

Connecting the previous data trends for new customers for giving them useful recommendations

Encouraging our customers to track their data for expanding the dataset

Correlate different data for better customer satisfaction

Dataset:

The dataset includes fitness tracking data collected from users doing various health and exercise activities. We will be using tools like Python (Pandas, NumPy), SQL, and BI apps such as Tableau.

Potential problems: Some entries may miss key details like workout duration or intensity. Users may log activities differently, leading to categorization challenges. We could also come across duplicated data.

KPIs:

- hydration/heart-rate ->Average heart rate based on the hydration level
- stress-level/sleep -> Average hours of sleep per stress level
- daily steps/stress-level ->Average stress level based on daily steps
- calories burned/activity-type->Average calories burned per activity type
- smoking/duration/intensity->Average exercise duration and intensity for smokers vs non-smokers
- daily steps/stress-level/age ->Average daily steps based on stress level and age
- intensity/calories burned->Average calories burned per intensity level
- weight/duration/activity-type->Average workout duration per weight category and activity type
- age/duration->Average workout duration based on age
- blood_pressure_systolic/stress/blood_pressure_diastolic->Average systolic and diastolic blood pressure per stress level

Part 1

Key Analytical Questions for Data Warehouse (DW & OLAP)

How do user workout trends (activity type) vary across different demographics (age, gender,BMI)?

What are the most effective activity types based on calories burned?

How does smoking affect health (stress level, sleep, duration and intensity of exercise)?

How demographic information can affect fitness performance (duration of exercise, steps count, intensity)?

What is the relationship between blood pressure(diastolic or systolic) and stress level?

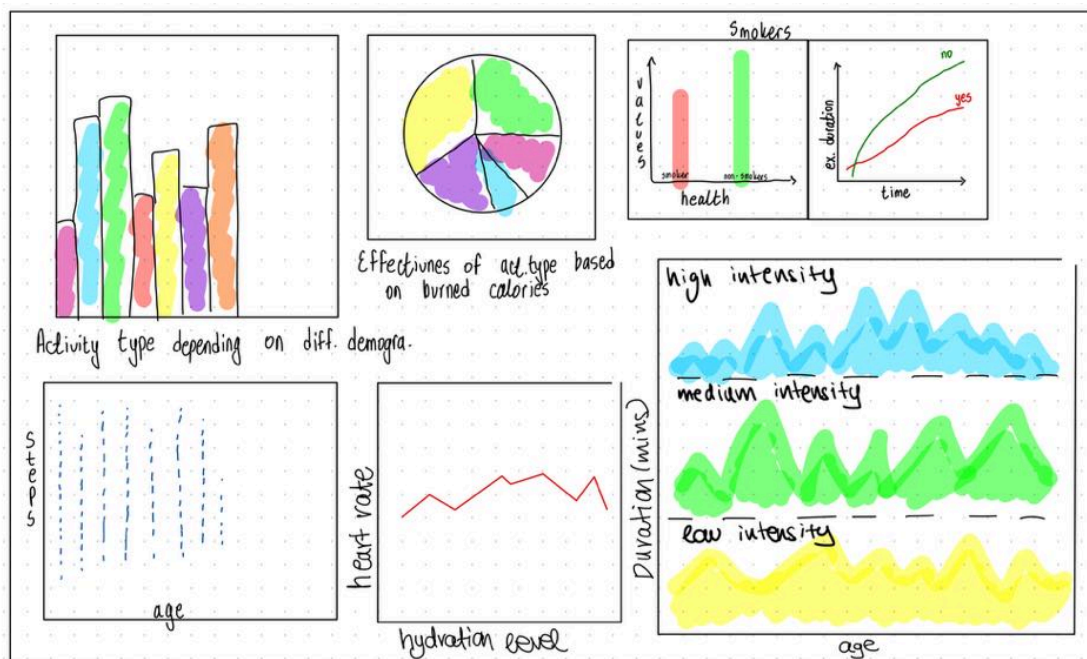
What is the relationship between BMI and physical fitness, considering factors as exercise duration and calories burned?

What is the average number of steps per stress level and age group?

What is the relationship between stress levels and sleep duration across different age groups?

How does hydration level affect the average heart rate during exercise?

2. Data Aggregation & Visualization Needs



3. OLAP Actions & Capabilities

To allow deep exploration of data, the BI solution will support:

- Drill-Down: View detailed workout statistics per user category (e.g., age groups).
- Roll-Up: Aggregate data for high-level overviews (e.g., national fitness trends).
- Slice and Dice: Filter workout data by demographic parameters or activity type.
- Pivoting: Change perspectives on data (e.g., see BMI effects on exercise duration vs. calories burned).

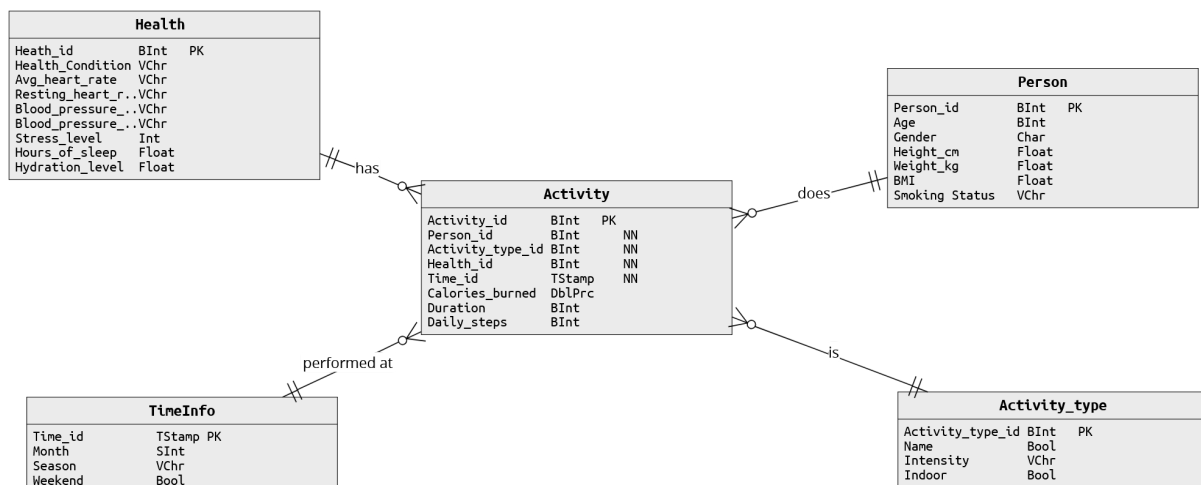
4. Mockups & User Interface Design

The BI system will include:

- User Overview Dashboard: Displays general health and fitness statistics.
- Workout Effectiveness Report: Shows the best exercise types based on calories burned.
- Demographic Fitness Insights: Analyzes how different demographics respond to fitness activities.
- Health Risk Analysis: Correlates health conditions with lifestyle habits (e.g., smoking impact on exercise).

1.2

Star schema:



1.3

1. Identifying “must-have” Requirements

a. ETL Tool Requirements (Pentaho)

- Visual Modeling & Process Flow Definition: Pentaho provides a graphical dragging and dropping interface for defining ETL workflows without requiring extensive coding.
- Data Integration: Supports multiple data sources (databases, APIs, CSV files) to ensure smooth data ingestion.
- Data Cleansing and Transformation: Handles missing values, duplicates and inconsistencies.
- Scalability and Automation: Supports scheduled job execution for continuous data updates.

b. Data Warehouse Size Considerations (PostgreSQL)

- The dataset contains various health and fitness metrics. While the exact size depends on user adoption, PostgreSQL is designed to handle large datasets efficiently.
- PostgreSQL supports indexing, partitioning and optimization techniques to ensure high performance even as the data grows.
- The dataset is connected to PostgreSQL via a Python script, written for that purpose.

c. OLAP Tool Requirements (Tableau)

- Web-Based Access: Tableau supports both desktop and web-based reporting, ensuring accessibility.
- Interactive Dashboards: Enables drill-down, filtering and real-time analysis of fitness data.
- Advanced Visualizations: Supports heatmaps, bar charts, line graphs and pivot tables to analyze trends effectively.

d. Licensing Considerations

- PostgreSQL: Open-source and free, ensuring cost-effectiveness.
- Pentaho: Offers an open-source community edition and a commercial version with extended features.
- Tableau: Commercial tool widely used for BI and analytics. A trial version is available and it is among the most widely adopted BI tools in the industry.

e. Additional Considerations

- Ease of Integration: All selected tools integrate well with each other ensuring a seamless data pipeline.
- Industry Adoption & Community Support: PostgreSQL, Pentaho and Tableau are well-documented and widely used, reducing implementation risks.

2. Criteria for Comparing Software Solutions

The software was introduced in practical class, and taking in consideration the great performance, ease of usage and scalability, that was the criteria for choosing it.

3. Approach for Searching and Comparing Multiple Solutions

Step 1: Define Requirements

We identified the key requirements for each tool category (Storage, ETL, OLAP) based on:

- Data size and structure
- Need for automation and scalability
- Visualization and reporting capabilities

Step 2: Market Research and Comparison

We researched available tools based on industry reports, user reviews and documentation.

- Storage: PostgreSQL, MySQL, Microsoft SQL Server
- ETL: Pentaho, Talend, Apache Nifi
- OLAP/BI: Tableau, Power BI

Step 3: Final Selection

PostgreSQL, Python, Pentaho and Tableau were selected based on their performance, scalability and industry adoption ensuring a reliable analytical solution for our fitness tracking project.

1.4

1. **Identify and describe the sources of data:** The data was taken from a Kaggle dataset named “FitLife Health & Fitness Tracking” and they describe all the activity of a person within the year 2024. More specifically, the database keeps a log of the demographic information of each person, activity metrics for each activity it participates in, health indicators and lifestyle metrics.
2. **Present the overall ETL plan for your solution:** For the ETL we used Pentaho alongside with some python script. More specifically, for the Extract phase we used a temporary Pandas Dataframe (staging area). For the Transformation phase we checked for missing values and for duplicate information through python to find out that our dataset did not have any such values so we continued with the rest of the data cleaning process. We standardized data related to heart rate and blood pressure, categorizing them to “HIGH/MEDIUM/LOW” according to their value (for average heart rate and resting heart rate ‘normal’ is between 60 to 100, for systolic blood pressure it is less than 120 mm Hg and for diastolic it is under 80 mm Hg) by using consecutive number range steps. Finally, , for the Load phase we moved the cleaned data into the Data Warehouse.
3. **Describe the staging area:** We used Pandas libraries in Python and steps Number Range in pentaho.
4. **For each significant action in the plan, explain why it is necessary and how it is implemented:**
Step 1- Extract Data: We loaded the raw data from the CSV file to be able to clean them and use them in our favor.
Step 2- Transform Data : We use the consecutive number range steps to categorize numeric values into meaningful groups.
Step 3- Load Data: We stored the transformed data into the Data Warehouse to save our cleaned data. We used Python to fill the Data Warehouse with data.
5. **Present the major challenges to the implementation:** One of the first challenges that we encountered was the data standardization of heart rates and blood pressures in groups through pentaho. Furthermore, we had performance issues because of the size of the dataset. Later, we had a problem connecting the data from Pentaho to PostgreSQL, which kept us stuck for a couple of days before moving on with the tableau.
6. **Present the following metrics:**

Size of Source Data: Approximately 78 MB.

Size of Data on First Load: 5 minutes

Size of Data on Subsequent Loads: 2.5 minutes

Time Elapsed on the First Load: 6 minutes

Time to fill every table with data: 3 minutes

7. Identify problems with the source and DW data that were not dealt with (if they exist):

1) The source database contained a column named health condition which used the word "None" to specify that the person had not a health condition. In the process of PostgreSQL, the system misinterpreted the word "None" as Null Value. As a result to avoid the issues we had to remove the column.

2) Some solvable issues with the Python script, regarding activity table

8. Explain how the ETL process is automated to allow future data updates in the DW and the update strategy for the dimensions, facts tables, and frequency:

The data can be frequently changed for the next year, when the data is collected.

9. Describe How the OLAP Data Is Presented to Final Users, How They Can Access It, and Modify the Search Parameter:

OLAP data is presented to the users through interactive dashboards which contain multiply visual elements as graphs and charts. These dashboards are designed to offer a user-friendly interface that allows users to explore and analyze data dynamically. Users can access OLAP data with BI tools and reporting platforms (as Tableau or Power BI) which are usually web-based or part of enterprise systems. All these tools offer an easy to use interface, allowing users to explore data without the need of SQL knowledge or technical expertise. Users are able to interact through already built dashboards which contain charts and graphs, allowing them to roll up or drill down into more details, drag and drop or even perform slice and dice. Also they can modify search parameters dynamically by adjusting filters, measures, marks, dimensions and group data in real time, creating their own graphs and visualizations based on their needs. This allows them to refine queries, compare different datasets and extract actionable information.

10. Describe the analyses being performed.

Workout Trends Across Demographics:

1) Examines how workout types vary based on age, gender, weight and bmi

2) Slice-Dice to compare different demographic groups

Activity Effectiveness Analysis:

1) Shows the most effective workout types based on the calories burned

2) Compare workout intensity with calories burned

Impact of Lifestyle Factors on Health Metrics:

1) Analyze how smoking affects stress, sleep, duration and intensity

2) Examines the relationship between hydration and average heart rate

Correlation between Health Metrics:

1) Examines blood pressure (systolic or diastolic) and stress level

2) Analyzes how calories burned connect with BMI and activity type

3) Investigates the relationship between stress levels and daily steps across different age groups

User Engagement and Step Count patterns:

1) Evaluates average steps count per stress level and age group

11. Present and discuss the initial findings.

Fitness apps can encourage users to stay active during weekends with motivational challenges.

Health instructors can tailor plans based on user fitness levels, promoting preventive healthcare strategies.

Gyms and fitness programs can design flexible workout schedules to match users' most active time periods.

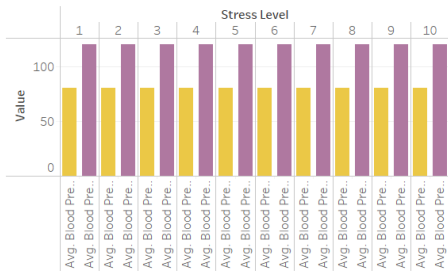
12. Discuss results from the business perspective and explain how the information can be used for Decision Support.

The dataset contains information about users' daily activities, calories burned, heart rate, and exercise patterns. By analyzing these data points, businesses can:

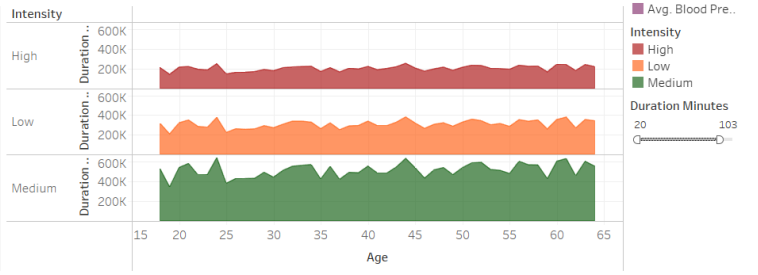
- Identify trends in user activity levels (e.g., peak exercise times, most common workout types).
- Assess the effectiveness of fitness programs by tracking improvements in users' health metrics over time.
- Prepare clients customized workout plans
- Target specific audience in different seasons
- Optimize memberships and enlarge customer network
- Develop smart programs/smart apps for customers



Blood pressure / Stress level



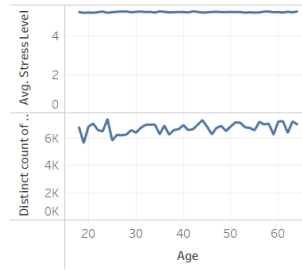
Impact of age on fitness duration



Activity type vs Calories burned



Steps per stress and age group



Hydration

